

SUNKU VENKATA REVANTH KUMAR

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PROFESSIONAL SUMMARY

Data & AI Engineer with **3+ years** building and shipping **production-grade AI systems on AWS and Azure**. Core focus on **RAG, multi-agent orchestration, and scalable cloud-native pipelines**. With equal weight on **correctness, cost efficiency, and operational reliability**. Reduced **~8k** monthly support tickets, cut cloud infrastructure spend by **~40%**, and compressed procurement cycle times from **3-7 days to <20 mins**. Comfortable owning a system end-to-end: from **RAG pipeline** design and LLM evaluation to **Docker deployment, CI/CD, & CloudWatch observability**.

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, SQL, LangChain, LangGraph, CrewAI, Pydantic, Streamlit

GenAI & LLMs: RAG, Multi-Agent Workflows, Prompt Engineering, LLM Evaluation, Hallucination Mitigation, Fine-Tuning, NLP, OpenAI GPT, Claude, Gemini, Hugging Face Transformers

Retrieval & Vector Databases: FAISS, ChromaDB, Semantic Retrieval, Hybrid Search, Cross-Encoder Re-ranking, Metadata Filtering, Query Expansion, Context Compression, Embedding Models (OpenAI, BGE)

Cloud, MLOps & Infrastructure: AWS (Lambda, Step Functions, Glue, S3, EC2, API Gateway, EventBridge, CloudWatch), Azure AI Foundry, Azure Cognitive Services, MLflow, CI/CD Pipelines, Docker

Governance & LLMops: RAGAS, Human-in-the-Loop (HITL), Guardrails, Auditability, Token Monitoring, Cost Optimization, Exception Handling, Compliance Monitoring, LLM Observability

Automation & Data Engineering: UiPath, ETL Pipelines, Workflow Automation, Queue Management, Web/Desktop Automation

WORK EXPERIENCE

Data & AI Engineer | Genpact (Client: GE Vernova)

Jul 2024 – Present | Bengaluru

- Built a **LangGraph Agentic Orchestration pipeline**(Extraction → Matching → Exception Handling) to automate invoice reconciliation against live vendor contracts and compliance documents
- Engineered a **hybrid RAG stack using FAISS** vector store with dense-sparse search and cross-encoder re-ranking across 10,000+ vendor contracts. Choose FAISS for deterministic latency SLAs and cost efficiency (**40% savings**); improved retrieval precision from **65% → 92%** MRR without hallucinated references.
- Built a **3-layer hybrid validator (Pydantic → deterministic rules → LLM)** at layer 2 for ~65% of invoices to avoid unnecessary LLM calls.
- Implemented configurable **exception-routing state machine (Approved/Warning/Critical/HITL)** with confidence scoring, achieving 85% STP. Eliminated **8K+** monthly support tickets..
- Deployed **RAGAS faithfulness + relevance eval** with **token/latency** dashboards compressed **approval cycles from 3–7 days to <20 minutes**; system auditable end-to-end.

Enterprise Contract Intelligence & RAG Platform

- Built **CrewAI**-based multi-agent system (**Writer, Analyst, Compliance, Risk**) on AWS with deterministic orchestration, structured output validation, and guardrails to auto-generate human-reviewable contract drafts while maintaining enterprise-safe response governance.
- Engineered **metadata-driven** retrieval pipeline combining embeddings, dense-sparse search, and cross-encoder re-ranking across dynamic knowledge bases (policies, SOPs, compliance records); improved grounding accuracy and response relevance **82% → 94%** through context-aware retrieval and query expansion strategies.
- Implemented **LLMops observability, metadata tagging, retrieval experimentation, and Streamlit interfaces** to optimize **RAG performance** and streamline low-level contract analysis workflows.

Enterprise SaaS Governance & AI-Powered Observability Platform

- Designed distributed ETL processing **500k+ records/4hrs** across multiple production clusters, fully containerized via **Docker** and deployed using automated **CI/CD pipelines**.
- Achieved automated license utilization tracking, reduced **annual SaaS spend by 40%** and eliminated access-related **admin tickets by 70%** through resilient data synchronization and **automated retry** mechanisms.
- Developed a **GenAI-driven log intelligence pipeline** leveraging open-source LLMs to convert massive CloudWatch log streams into contextual root-cause reports and **stakeholder-ready insights** after every run, significantly reducing manual operational analysis beyond conventional regex-based systems.

Software Engineer Specialist | GE India

Jan 2023 – Jun 2024 | Bengaluru

AWS ML Inference & Monitoring Platform

- Architected a **centralized AWS-based ML inference infrastructure** to consolidate distributed team-specific ML workloads, scalable model deployment, orchestration and **centralized compliance monitoring** across enterprise environments.
- Developed **FastAPI-based inference workflows** integrated with **AWS API Gateway** to standardize model serving, input/output validation, and **API-driven ML execution** for multiple business teams on a shared cloud infrastructure.
- Established centralized **MLOps observability and resource utilization tracking**, saving annual cloud infrastructure costs by **~40%** and improving overall **cloud utilization to ~90%** through workload consolidation and optimized compute allocation.

Enterprise RPA | Cross-System Data Sync Bot

- Built UiPath automations for complex ETL across CSV/Excel files of **500k+ records** and web portals, eliminating **120K+** manual work hours.

EDUCATION & Certifications

- B.Tech, Computer Science & Engineering** - Vellore Institute of Technology (CGPA: 8.2/10) 2019 - 2023
- Certifications:** [Advanced Gen AI \(Genpact AI Giga Academy\)](#) | [Azure Fundamentals \(AZ-900\)](#) | [AWS Academy Cloud Practitioner](#)